

Lucie Choi

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Education

University of California, Berkeley

Bachelor of Arts in Computer Science (Focus in Systems) | GPA: 3.65

Berkeley, CA

2016 – 2020

Stanford University

Non-Degree Option (NDO), Computer Science (Focus in Visual Computing)

Stanford, CA

2025 – 2026

- **Coursework:** CS 348C (Advanced Animation and Simulation), CS 448Z (Physically Based Sound), CS 248B (Character Animation and Simulation), CS 248A (Rendering, Geometry, and Image Manipulation), CS 148 (Intro to Computer Graphics), EE 263 (Linear Dynamical Systems)

Professional Experience

Google

Senior Software Engineer, Android XR Graphics

San Jose, CA

Dec 2021 – Present

- **XR Glasses Graphics (2026):** Architecting low-latency, power-efficient Android graphics and display pipelines in XR glasses. Integrating Qualcomm hardware-accelerated Late Stage Reprojection (LSR) to minimize motion-to-photon latency.
- **Shader Compilers & Vulkan Backend (2025–2026):** Contributed to open-source **DXC** and **LLVM** compilers for HLSL-to-SPIR-V translation, implementing **CombinedImageSampler** support and GPU reconvergence features for Vulkan.
- **Likeness Photorealistic Avatars (Tech Lead, 2024–2025):** Led end-to-end service architecture and mobile app development for 3D photorealistic avatar creation and data streaming between 1st party VR headsets and Android phones. [Press Coverage](#)
- **YouTube VR (Tech Lead, 2023–2024):** Engineered real-time video co-watching engine for Meta Quest headsets, including playback synchronization, party management, and data encryption. [Press Coverage](#)
- **Document AI Core Infrastructure (2021–2023):** Developed backend services to allow enterprise customers to manage domain-specific documents and run various machine learning tasks, such as OCR, parsing, and classification.

Intuit

Software Engineer, Full-Stack

Mountain View, CA

Jun 2020 – Oct 2021

- Designed backend microservices and data migration architecture to securely transition the acquisition methods for customers' banking data.

UC Berkeley RISELab

Undergraduate Research Assistant, Database Systems

Berkeley, CA

Jan 2019 – Dec 2019

- Co-developed **DIEL**, a declarative framework unifying UI interactions as event tables via SQL. Built a cross-node query compiler and runtime that optimizes dynamic view materialization across client (SQLite/sql.js) and server (PostgreSQL) backends.

Selected Projects

Cloth Simulation: Position-Based Dynamics

Stanford CS 348C

- Implemented a custom Position-Based Dynamics (PBD) cloth solver from scratch in Python/NumPy featuring stretch, bend, and shear constraints, spatial hashing for self-collisions, and object collision, rendered in Houdini.

Acoustic Peace Transfer: Modal Sound Synthesis & Radiation Transfer

Stanford CS 448Z

- Simulated modal acoustics and acoustic radiation transfer of the Bell of King Seongdeok using Linear Modal Analysis, Equivalent Source Method (ESM), and audio-driven particle visualization in SciPy and Houdini.

Selected Skills

Languages: C++, Python

Graphics APIs & Systems: Vulkan, OpenGL, HLSL, Android Graphics (AOSP), Taichi Lang, ThreeJS

DCC Tools: Houdini, Maya

Leadership & Community

Google Pride Korea

Committee Member

Remote / Seoul, South Korea

2026 – Present

- Organize the company's various LGBTQ+ initiatives in South Korea—including Seoul Queer Culture Festival, Pride Parade Walk, and in-company events—championing inclusion within Google while inspiring greater diversity across Korean society at large.